P1 $\psi(t=0) = \frac{4}{5}\psi_{100} + \frac{3i}{5}\psi_{211}$

By construction, $\hat{H}\psi_{nlm} = E_n \psi_{nlm}$, $E_n = -\frac{E_1}{n^2}$, $n=1,2,3\ldots$

(a) $\langle E\rangle = \langle\psi|\hat{H}|\psi\rangle = \left(\frac{4}{5}\psi_{100}^* - \frac{3i}{5}\psi_{211}^*\right)\hat{H}\left(\frac{4}{5}\psi_{100} + \frac{3i}{5}\psi_{211}\right)$

$= \left(\frac{4}{5}\psi_{100}^* - \frac{3i}{5}\psi_{211}^*\right)\left(\frac{4}{5}E_1\psi_{100} + \frac{3i}{5}E_2\psi_{211}\right) =$

$= \frac{16}{25}E_1\langle\psi_{100}|\psi_{100}\rangle + \frac{9}{25}E_2\langle\psi_{211}|\psi_{211}\rangle =$

$= \frac{16}{25}E_1 + \frac{9}{25}\frac{E_1}{4} = \frac{E_1}{25}\left(16 + \frac{9}{4}\right) = \frac{73}{100}E_1.$

$\langle L_z\rangle = \langle\psi|\frac{4}{5}\hbar\cdot 0\,\psi_{100} + \frac{3i}{5}\hbar\cdot 1\,\psi_{211}\rangle =$

$= \frac{9}{25}\hbar\langle\psi_{211}|\psi_{211}\rangle = \frac{9\hbar}{25}$

$\langle \vec{L}^2\rangle = \langle\psi|\frac{4}{5}\hbar^2 0(0+1)\psi_{100} + \frac{3i}{5}\hbar^2 1(1+1)\psi_{211}\rangle$

$= \frac{9}{25}\hbar^2\cdot 2\,\langle\psi_{211}|\psi_{211}\rangle = \frac{18}{25}\hbar^2.$

This is because ψ_{nlm} are eigenstates of energy (E_n) angular momentum $L^2\psi_{nlm} = \hbar^2 l(l+1)\psi_{nlm}$ and its z-component $L_z\psi_{nlm} = \hbar\cdot m\,\psi_{nlm}$.

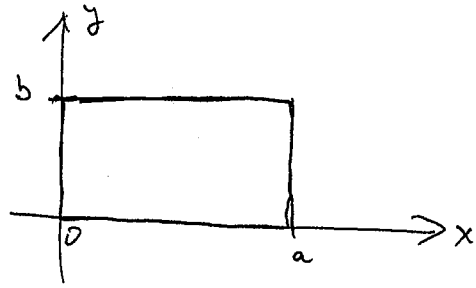
(b) $\psi(t) = \frac{4}{5}\psi_{100}e^{-iE_1t/\hbar} + \frac{3i}{5}\psi_{211}e^{-iE_2t/\hbar}$

$= \frac{4}{5}\psi_{100}e^{-iE_1t/\hbar} + \frac{3i}{5}\psi_{211}e^{-iE_1t/4\hbar}.$

None of the expectation values in (a) depends on time because $\langle\psi_{n'l'm'}|\psi_{nlm}\rangle = \delta_{nn'}\delta_{ll'}\delta_{mm'}$ always, for any t.

P2

$$V(x,y) = \begin{cases} 0 & , \ 0 \le x \le a \text{ and } 0 \le y \le b \\ \infty & , \text{ otherwise} \end{cases}$$



$$\Rightarrow \ H = \begin{cases} \frac{p_x^2}{2m} + \frac{p_y^2}{2m} & , \ 0 \le x \le a \text{ and } 0 \le y \le b \\ \infty & , \text{ otherwise} \end{cases}$$

Thus wavefunction must be zero outside the rectangle.

(a) Inside rectangle

$$-\frac{\hbar^2}{2m}\frac{\partial^2 \psi}{\partial x^2} - \frac{\hbar^2}{2m}\frac{\partial^2 \psi}{\partial y^2} = E\psi \quad (1)$$

let $\psi(x,y) = X(x)\,Y(y)$: plug in (1) and divide by XY

$$\Rightarrow \ \underbrace{-\frac{\hbar^2}{2m}\frac{1}{X}\frac{d^2X}{dx^2}}_{E_x} \underbrace{-\frac{\hbar^2}{2m}\frac{1}{Y}\frac{d^2Y}{dy^2}}_{E_y} = E \quad , \ E = E_x + E_y$$

E_x (E_y) are eigenvalues of one-dimensional problem

with $E_x = \frac{\pi^2\hbar^2}{2ma^2} n_x^2 \quad \left(E_y = \frac{\pi^2\hbar^2}{2mb^2} n_y^2 \right)$

$n_x = 1, 2, 3 \ldots \qquad n_y = 1, 2, 3 \ldots$

$$\psi(x,y) = \sqrt{\frac{2}{a}} \sin\left(\frac{\pi n_x}{a} x\right) \sqrt{\frac{2}{b}} \sin\left(\frac{\pi n_y}{b} y\right)$$

(b) $$E_{n_x n_y} = \frac{\pi^2\hbar^2}{2m}\left(\frac{n_x^2}{a^2} + \frac{n_y^2}{b^2}\right)$$

Ground state (lowest energy): $n_x = n_y = 1 \Rightarrow E_{11} = \frac{\pi^2\hbar^2}{2m}\left(\frac{1}{a^2} + \frac{1}{b^2}\right)$

The first excited state : $n_x = 2$, $n_y = 1$

$\hookrightarrow E_{21} = \frac{\pi^2\hbar^2}{2m}\left(\frac{4}{a^2} + \frac{1}{b^2}\right)$

Note that since $a > b$ as given, $E_{12} = \frac{\pi^2\hbar^2}{2m}\left(\frac{1}{a^2} + \frac{4}{b^2}\right) > E_{21}$

$\Downarrow$

$\frac{1}{a^2} < \frac{1}{b^2}$

The second excited state : there are several possibilities (depending on $\frac{a}{b}$)

(1) $n_x = 1$, $n_y = 2$

$E_{12} = \frac{\pi^2\hbar^2}{2m}\left(\frac{1}{a^2} + \frac{4}{b^2}\right)$

OR

(2) $n_x = 3$, $n_y = 1$

$E_{31} = \frac{\pi^2\hbar^2}{2m}\left(\frac{9}{a^2} + \frac{1}{b^2}\right)$

OR (3) $n_x = 2$, $n_y = 2$

$E_{22} = \frac{\pi^2\hbar^2}{2m}\left(\frac{4}{a^2} + \frac{4}{b^2}\right)$

[$E_{22} > E_{12}$ always
and $E_{31} < E_{22}$ for $\frac{a}{b} > \sqrt{\frac{5}{3}}$

$E_{12} < E_{31}$ when $\frac{1}{a^2} + \frac{4}{b^2} < \frac{9}{a^2} + \frac{1}{b^2}$

$\frac{3}{b^2} < \frac{8}{a^2} \Rightarrow \frac{a}{b} < \sqrt{\frac{8}{3}}$

Thus $\begin{cases} \text{if } 1 < \frac{a}{b} < \sqrt{\frac{8}{3}}, \text{ then } E_{12} \text{ is second excited state} \\ \text{if } \frac{a}{b} > \sqrt{\frac{8}{3}}, \text{ then } E_{31} \text{ is second excited state.} \end{cases}$

part (d) answer

All states are non-degenerate (meaning, a given allowed energy is uniquely determined by a pair (n_x, n_y).)

(c) If $a = b$, then E_{11} is non-degenerate. (ground state)

First excited $E_{21} = E_{12} = \frac{\pi^2\hbar^2}{2m}\frac{5}{a^2}$ is 2-fold degenerate : $\begin{cases} n_x = 2, n_y = 1 \\ n_x = 1, n_y = 2 \end{cases}$

Second excited $E_{22} = \frac{\pi^2\hbar^2}{2m}\frac{8}{a^2}$ is : $n_x = 2, n_y = 2$

(d) $\langle r^2 \rangle = \langle x^2 + y^2 \rangle = \int (x^2+y^2)\, X^2(x)\, Y^2(y)\, dx\, dy =$

$$= \int x^2 X^2(x)\, dx \cdot \underbrace{\int Y^2(y)\, dy}_{1} + \underbrace{\int X^2(x) dx}_{1} \int y^2 Y^2(y) dy$$

$$= \langle x^2 \rangle_x + \langle y^2 \rangle_y$$

Where $\langle x^2 \rangle_x$ and $\langle y^2 \rangle_y$ are determined by the $X(Y)$ one-dimensional wave functions!

$$\Rightarrow \langle x^2 \rangle_x = \frac{2}{a} \int_0^a dx\, x^2 \underbrace{\sin^2\left(\frac{\pi n_x}{a} x\right)}_{\frac{1}{2}\left(1 - \cos\left(\frac{2\pi n_x}{a} x\right)\right)} =$$

$$= \frac{1}{a} \int_0^a x^2 \left(1 - \cos\left(\frac{2\pi n_x}{a} x\right)\right) dx$$

To calculate $I = \int_0^a -x^2 \cos\left(\frac{2\pi n_x}{a} x\right) dx$ we note that

$$I = \frac{d^2}{d\beta^2} \int_0^a \cos(\beta x)\, dx \Big|_{\beta = \frac{2\pi n_x}{a}} = \frac{d^2}{d\beta^2} \frac{\sin \beta x}{\beta} \Big|_0^a =$$

$$= \frac{d^2}{d\beta^2} \left(\frac{\sin \beta a}{\beta} - 0 \right) = \frac{d}{d\beta} \left(\frac{a \cos \beta a}{\beta} - \frac{\sin \beta a}{\beta^2} \right) =$$

$$= \left(-a^2 \frac{\sin \beta a}{\beta} - \frac{2a \cos \beta a}{\beta^2} + \frac{2 \sin \beta a}{\beta^3} \right) \Big|_{\beta = \frac{2\pi n_x}{a}} = -2a \frac{\cos(2\pi n_x)}{\left(\frac{2\pi n_x}{a}\right)^2} =$$

$$= -\frac{2a^3}{4\pi^2 n_x^2}$$

Thus $\langle x^2 \rangle = \frac{1}{a} \left(\frac{a^3}{3} - \frac{2a^3}{4\pi^2 n_x^2} \right) = a^2 \left(\frac{1}{3} - \frac{1}{2\pi^2 n_x^2} \right)$

Finally, $\langle r^2 \rangle = a^2 \left(\frac{1}{3} - \frac{1}{2\pi^2 n_x^2} \right) + b^2 \left(\frac{1}{3} - \frac{1}{2\pi^2 n_y^2} \right)$.

P3 $H^0 = \frac{p^2}{2m} + \frac{1}{2} m\omega^2 x^2$, $E_n^0 = \hbar\omega(n+\frac{1}{2})$, $\psi_n^0 = |n\rangle$

(a) $H'_\alpha = \alpha x^3$

$$x = \sqrt{\frac{\hbar}{2m\omega}}\,(a_+ + a_-)$$

$$\Rightarrow \quad \langle 0|x^3|0\rangle = 0 \quad \text{[odd function under the integral.]}$$

But $E_0^1 = \langle 0|H'_\alpha|0\rangle = \alpha\langle 0|x^3|0\rangle = 0$

$$E_0^2 = \sum_{m\neq 0}{}' \frac{|\langle m|H'|0\rangle|^2}{E_0^0 - E_m^0} = -\frac{\alpha^2}{\hbar\omega}\sum_{m\neq 0}{}' \frac{|\langle m|x^3|0\rangle|^2}{+m}$$

Consider $x^3|0\rangle = \left(\frac{\hbar}{2m\omega}\right)^{3/2}(a_+ + a_-)(a_+ + a_-)(a_+ + a_-)|0\rangle =$

$$= \left(\frac{\hbar}{2m\omega}\right)^{3/2}(a_+ + a_-)(a_+ + a_-)\,|1\rangle = \left(\frac{\hbar}{2m\omega}\right)^{3/2}(a_+ + a_-)\left(\sqrt{2}\,|2\rangle + |0\rangle\right)$$

$$= \left(\frac{\hbar}{2m\omega}\right)^{3/2}\left(\sqrt{2}\cdot\sqrt{3}\,|3\rangle + |1\rangle + \sqrt{2}\cdot\sqrt{2}\,|1\rangle + 0\right)$$

$$= \left(\frac{\hbar}{2m\omega}\right)^{3/2}\left(\sqrt{6}\,|3\rangle + 3|1\rangle\right)$$

$$\Rightarrow \quad \langle m|x^3|0\rangle = \left(\frac{\hbar}{2m\omega}\right)^{3/2}\left(\sqrt{6}\,\delta_{m,3} + 3\delta_{m,1}\right)$$

Hence $E_0^2 = -\frac{\alpha^2}{\hbar\omega}\left(\frac{\hbar}{2m\omega}\right)^3 \sum_m{}' \left\{\frac{(\sqrt{6}\,\delta_{m,3})^2}{3} + \frac{(3\delta_{m,1})^2}{1}\right\} =$

$$\frac{6}{3} + 9 = 11$$

$$= -\frac{\alpha^2}{\hbar\omega}\left(\frac{\hbar}{m\omega}\right)^3 \frac{11}{8}.$$

(b) $H'_\beta = \beta x^4$

$$E_0^1 = \beta\langle 0|x^4|0\rangle$$

But, using (a), $x^4|0\rangle = \left(\frac{\hbar}{2m\omega}\right)^2(a_+ + a_-)\left(\sqrt{6}\,|3\rangle + 3|1\rangle\right) =$

$$= \left(\frac{\hbar}{2m\omega}\right)^2 \left(\sqrt{6}\cdot\sqrt{4}\,|4\rangle + 3\cdot\sqrt{2}\,|2\rangle + \sqrt{6}\cdot\sqrt{3}\,|2\rangle + 3|0\rangle\right)$$

$$\Rightarrow \langle 0|x^4|0\rangle = \left(\frac{\hbar}{2m\omega}\right)^2 3\langle 0|0\rangle = \frac{3}{4}\left(\frac{\hbar}{m\omega}\right)^2$$

$$\Rightarrow E_0^1 = \frac{3}{4}\beta\left(\frac{\hbar}{m\omega}\right)^2.$$

(c) For general state $|n\rangle$:

$$E_n^1 = \langle n|\beta x^4|n\rangle = \frac{3}{4}(1+2n+2n^2)\beta\left(\frac{\hbar}{m\omega}\right)^2.$$

Work it out ~~by~~ yourselves!

P4

$$S_z = \hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} \qquad S_x = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \qquad S_y = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix}$$

(a)
$$[S_y, S_z] = \frac{\hbar^2}{\sqrt{2}} \left\{ \underbrace{\begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}}_{\begin{pmatrix} 0 & 0 & 0 \\ i & 0 & i \\ 0 & 0 & 0 \end{pmatrix}} - \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix}}_{\begin{pmatrix} 0 & -i & 0 \\ 0 & 0 & 0 \\ 0 & -i & 0 \end{pmatrix}} \right\}$$

$$= \frac{\hbar^2}{\sqrt{2}} \begin{pmatrix} 0 & i & 0 \\ i & 0 & i \\ 0 & i & 0 \end{pmatrix} = i\hbar \cdot \underbrace{\frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}}_{S_x}$$

as it should be!

(b)
$$S_x^2 + S_y^2 + S_z^2 = \hbar^2 \, 1(1+1) \cdot \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = 2\hbar^2.$$

matrix unity.

Indeed,
$$S_x^2 = \frac{\hbar^2}{2} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} = \frac{\hbar^2}{2} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

$$S_y^2 = \frac{\hbar^2}{2} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix} = \frac{\hbar^2}{2} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 2 & 0 \\ -1 & 0 & 1 \end{pmatrix}$$

$$S_z^2 = \hbar^2 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} = \hbar^2 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\Rightarrow S_x^2 + S_y^2 + S_z^2 = \hbar^2 \begin{pmatrix} \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 1 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} \end{pmatrix} + \hbar^2 \begin{pmatrix} \frac{1}{2} & 0 & -\frac{1}{2} \\ 0 & 1 & 0 \\ -\frac{1}{2} & 0 & \frac{1}{2} \end{pmatrix}$$

$$+ \hbar^2 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \hbar^2 \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} = 2\hbar^2 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

(c)
$$H = \frac{a}{\hbar^2} S_z^2 + \frac{b}{\hbar^2}\left(S_x^2 - S_y^2\right) = a\begin{pmatrix}1&0&0\\0&0&0\\0&0&1\end{pmatrix} +$$

$$+b\left\{\begin{pmatrix}\frac12&0&\frac12\\0&1&0\\\frac12&0&\frac12\end{pmatrix} - \begin{pmatrix}\frac12&0&-\frac12\\0&1&0\\-\frac12&0&\frac12\end{pmatrix}\right\} =$$

$$= a\begin{pmatrix}1&0&0\\0&0&0\\0&0&1\end{pmatrix} + b\begin{pmatrix}0&0&1\\0&0&0\\1&0&0\end{pmatrix} = \begin{pmatrix}a&0&b\\0&0&0\\b&0&a\end{pmatrix}$$

Eigenvalues: $\det\left(H - \lambda \hat{1}\right) = 0$

$$\Rightarrow \det\begin{pmatrix}a-\lambda&0&b\\0&-\lambda&0\\b&0&a-\lambda\end{pmatrix} = (a-\lambda)\begin{vmatrix}-\lambda&0\\0&a-\lambda\end{vmatrix} + b\begin{vmatrix}0&-\lambda\\b&0\end{vmatrix} =$$

$$= -\lambda(a-\lambda)^2 + b^2\lambda = \lambda\left(b^2 - (a-\lambda)^2\right) = 0$$

$$\Rightarrow \lambda_1 = 0, \quad \pm b = a - \lambda \Rightarrow \lambda_{2,3} = a \pm b$$

$$\Rightarrow \lambda_1 = 0, \lambda_2 = a+b, \lambda_3 = a-b$$

(d) Eigenstates

$$\begin{pmatrix}a&0&b\\0&0&0\\b&0&a\end{pmatrix}\begin{pmatrix}\alpha_1\\\alpha_2\\\alpha_3\end{pmatrix} = \lambda\begin{pmatrix}\alpha_1\\\alpha_2\\\alpha_3\end{pmatrix} \Rightarrow \begin{cases} a\alpha_1 + b\alpha_3 = \lambda\alpha_1 \\ 0 = \lambda\alpha_2 \\ b\alpha_1 + a\alpha_3 = \lambda\alpha_3 \end{cases}$$

$\underline{\lambda_1 = 0}$
$$\left.\begin{aligned} a\alpha_1 &= -b\alpha_3 \\ b\alpha_1 &= -\alpha\alpha_3 \end{aligned}\right\} \Rightarrow a\alpha_1 = +\frac{b^2}{\alpha}\alpha_1 \Rightarrow \begin{cases}\alpha_1 = 0\\ \alpha_3 = 0\end{cases}$$

$$\Rightarrow \psi_1 = \begin{pmatrix}0\\\alpha_2\\0\end{pmatrix} \Rightarrow \text{normalization requires } \alpha_2 = 1 : \psi_1 = \begin{pmatrix}0\\1\\0\end{pmatrix}$$

$\underline{\lambda_2 = a+b}$
$$a\alpha_1 + b\alpha_3 = (a+b)\alpha_1 \Rightarrow \alpha_3 = \alpha_1$$

$$\psi_2 = \begin{pmatrix}\alpha_1\\0=\alpha_2\\\alpha_1\end{pmatrix} \Rightarrow \text{normalizing: } \psi_2 = \frac{1}{\sqrt2}\begin{pmatrix}1\\0\\1\end{pmatrix}.$$

$\underline{\lambda_3 = a-b}$ $\quad \left.\begin{aligned} a\alpha_1 + b\alpha_3 &= (a-b)\alpha_1 \\ 0 &= (a-b)\alpha_2 \end{aligned}\right\} \Rightarrow \begin{aligned} \alpha_3 &= -\alpha_1 \\ \alpha_2 &= 0 \end{aligned}$

$$\psi_3 = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}.$$

(e) $\psi(0) = \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \sum_{i=1}^{3} r_i \psi_i$

where $r_i = \langle \psi_i | \psi(0) \rangle$

$$r_1 = (0, 1, 0)\begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = c_2$$

$$r_2 = \frac{1}{\sqrt{2}}(1, 0, 1)\begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \frac{c_1 + c_3}{\sqrt{2}}$$

$$r_3 = \frac{1}{\sqrt{2}}(1, 0, -1)\begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \frac{c_1 - c_3}{\sqrt{2}}.$$

$$\psi(0) = c_2 \psi_1 + \frac{c_1 + c_3}{\sqrt{2}} \psi_2 + \frac{c_1 - c_3}{\sqrt{2}} \psi_3 =$$

$$= c_2 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + \frac{c_1 + c_3}{\sqrt{2}} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + \frac{c_1 - c_3}{\sqrt{2}} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} =$$

$$= c_2 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + \frac{c_1 + c_3}{2} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + \frac{c_1 - c_3}{2} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}.$$

Thus $\psi(t) = \begin{pmatrix} 0 \\ c_2 \\ 0 \end{pmatrix} + \frac{c_1 + c_3}{2} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} e^{-i(a+b)t/\hbar} +$

$$+ \frac{c_1 - c_3}{2} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} e^{-i(a-b)t/\hbar}.$$

Can simplify a little bit more:

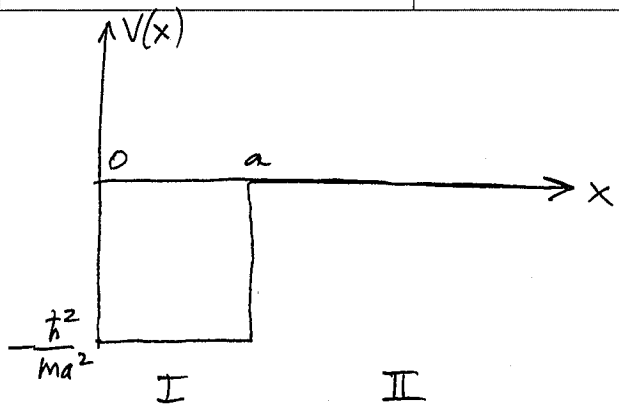
$$\frac{1}{2} e^{-iat/\hbar} \left\{ \begin{pmatrix} (c_1 + c_3)\, e^{-ibt/\hbar} \\ 0 \\ (c_1 + c_3)\, e^{-ibt/\hbar} \end{pmatrix} + \begin{pmatrix} (c_1 - c_3)\, e^{ibt/\hbar} \\ 0 \\ -(c_1 - c_3)\, e^{ibt/\hbar} \end{pmatrix} \right\} =$$

$$= \frac{1}{2} e^{-iat/\hbar} \begin{pmatrix} 2 c_1 \cos(bt/\hbar) - 2i\, c_3 \sin(bt/\hbar) \\ 0 \\ -2i\, c_1 \sin(bt/\hbar) + 2 c_3 \cos(bt/\hbar) \end{pmatrix}$$

$$\Rightarrow \psi(t) = \begin{pmatrix} e^{-iat/\hbar} \left[c_1 \cos(bt/\hbar) - i\, c_3 \sin(bt/\hbar) \right] \\ c_2 \\ e^{-iat/\hbar} \left[-i\, c_1 \sin(bt/\hbar) + c_3 \cos(bt/\hbar) \right] \end{pmatrix}.$$

<u>Check:</u> $\psi(t) \xrightarrow[t \to 0]{} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix}.$

P5



$E = -|E| < 0$

(a) $\psi(x) = 0$ for $x < 0$.

I: $\underline{0 \le x \le a}$ $\quad -\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V_0\psi = -|E|\psi = E\psi$

$$\frac{d^2\psi}{dx^2} = -\frac{2m}{\hbar^2}(E - V_0)\psi$$

Note that $E > V_0 > 0$ because <u>always</u> $E > \min(V(x))$.

$$\Rightarrow \frac{d^2\psi}{dx^2} = -k^2\psi \quad , \; k^2 = \frac{2m}{\hbar^2}(E - V_0)$$

II: $\underline{x > a}$ $\quad -\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = E\psi = -|E|\psi$

$$\frac{d^2\psi}{dx^2} = \frac{2m|E|}{\hbar^2}\psi = q^2\psi$$

$$q = \sqrt{\frac{2m|E|}{\hbar^2}} = \sqrt{-\frac{2mE}{\hbar^2}}$$

Thus $\begin{cases} \psi_I = A\cos kx + B\sin kx \\ \psi_{II} = C e^{-qx} \quad \text{(because } De^{qx} \text{ is not normalizable!)} \end{cases}$

(b) <u>At $x=0$</u> $\quad \psi_I(0) = 0 \Rightarrow A = 0$

$$\Rightarrow \psi_I(x) = B\sin kx$$

<u>At $x=a$</u> $\quad B\sin ka = Ce^{-qa} \quad (1)$

also, $\psi_I'(a) = \psi_{II}'(a) \Rightarrow Bk\cos(ka) = -Cqe^{-qa} \quad (2)$

Divide (2) by (1):

$$\kappa \frac{\cos(\kappa a)}{\sin(\kappa a)} = -q \Rightarrow \frac{\kappa}{q} = -\frac{\sin(\kappa a)}{\cos(\kappa a)} = -\tan(\kappa a)$$

Thus $\tan(\kappa a) = -\frac{\kappa a}{q a}$ (3)

Note that $\kappa^2 + q^2 = \frac{2m}{\hbar^2}(E - V_0 - E) = -\frac{2mV_0}{\hbar^2}$

$$\kappa^2 + q^2 = -\frac{2m}{\hbar^2} \cdot \frac{-\hbar^2}{ma^2} = \frac{2}{a^2}$$

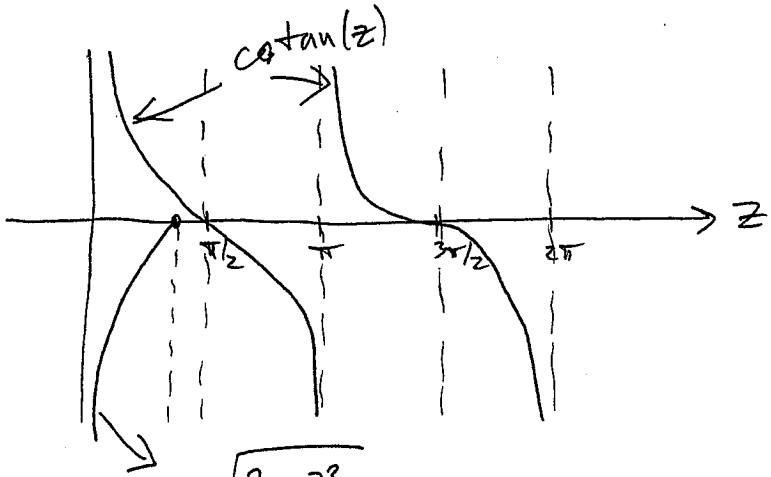
$$\Rightarrow (qa)^2 = 2 - (\kappa a)^2$$

Re-write (3) as: $\cot\text{an}(\kappa a) = -\frac{qa}{\kappa a}$

Introduce $z = \kappa a$: $q = \sqrt{2 - z^2}$ but q is real $\Rightarrow q = 0$ for $z \geq \sqrt{2}$

$$\frac{\cos(z)}{\sin(z)} = \cot\text{an}(z) = -\frac{\sqrt{2-z^2}}{z}$$

(c)



Important:
$\sqrt{2} \approx 1.41 < \frac{\pi}{2} \approx 1.5\ldots$

$-\frac{\sqrt{2-z^2}}{z} = f_1(z)$. Note that $f_1(z) = \begin{cases} 0 & \text{at } z = \sqrt{2} < \frac{\pi}{2} \\ -\infty & \text{at } z = 0 \end{cases}$

Thus $\cot\text{an}(z)$ and $f_1(z) = -\frac{\sqrt{2-z^2}}{z}$ do not intersect:

there are no bound states in this case!

(Because $V(x) \neq V(-x)$: asymmetric potential!)